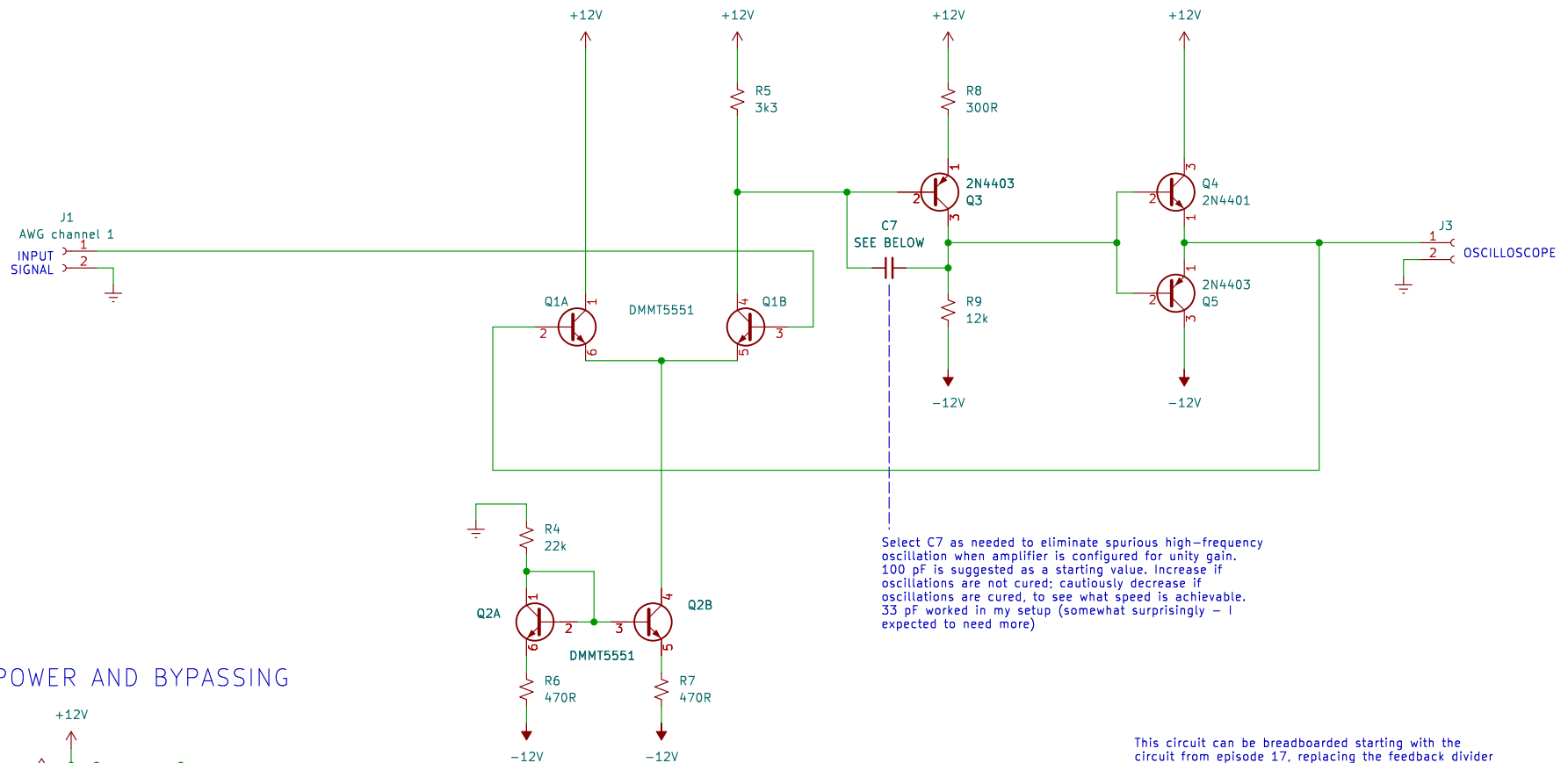


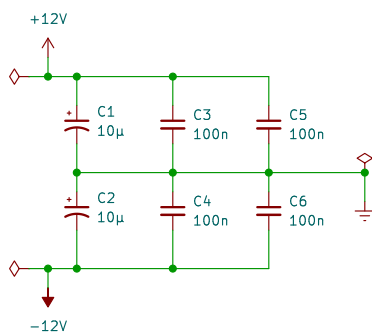
LONG-TAILED PAIR

GAIN STAGE

FOLLOWER STAGE

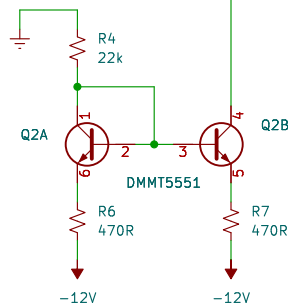


POWER AND BYPASSING



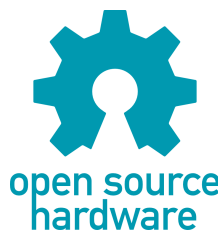
Place C3, C4 as close to Q1 as possible.
Place C5, C6 as close to Q4 and Q5 as possible.

500 μ A CURRENT SOURCE



This circuit can be breadboarded starting with the circuit from episode 17, replacing the feedback divider with a simple wire.

1. Make the modification, observe unity gain but likely high-frequency oscillation.
2. Attempt to cure the oscillation by adding compensation capacitor C7.



Transistors 101, episode 18

The differential pair becomes an op-amp

Kludges from Kevin's Cave

Sheet: /

File: LongTailedPair5.kicad_sch

Title: Long-tailed pair: the basic circuit

Size: USLetter Date: 2026-05-13

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